



VALUE ESTIMATE · V0.2.0 + V0.3.0

What the PortalX hardening is likely worth

A cost-avoidance **estimate model** for the fixes and UX work in PortalX for Android v0.2.0 and v0.3.0 — with every assumption visible and adjustable. There is no usage data yet, so nothing here is a measured saving.

ESTIMATED FIRST-YEAR COST AVOIDED · 50 USERS (PLACEHOLDER)

₹21k – ₹15.3 L

Expected ₹2.4 L (\$2,472) · range \$215 – \$15,764 · ₹96.9/\$

An estimate, built so you can replace every number

We do not have analytics, crash reports or headcount for PortalX, so this report does not claim money was saved. It estimates the **cost the v0.2.0 and v0.3.0 work is likely to avoid**, from public benchmarks where they exist and clearly labelled assumptions where they don't.

LOW

₹21k

\$215 · first year

EXPECTED

₹2.4 L

\$2,472 · first year

HIGH

₹15.3 L

\$15,764 · first year

- **Recurring, per year:** ₹8k / **₹68k** / ₹5.6 L — staff time, crashes, data fixes, security exposure. That is about **₹1,350 per user per year** (expected), so for any headcount N: recurring $\approx N \times ₹1,350$ (low ₹159, high ₹11,201).
- **One-time:** ₹13k / **₹1.7 L** / ₹9.7 L — engineering effort not spent hot-fixing 43 defects after release. It does not grow with headcount.
- **Biggest levers:** the one-time engineering avoidance (expected), then UX time savings and security exposure; at higher headcounts the per-user items dominate (sensitivity chart, p. 4).
- **Key assumptions:** 50 users (placeholder), loaded cost ₹250/h (low ₹130/h from PLFS 2025 ^[2]), seconds saved per flow per day, production-fix multiplier 3× (contested ^[3]), ₹6,456 per breached record (IBM 2026 India ^[1]), 15.4% quit after a crash ^[4].

How to read this. Low assumes PLFS-level wages and small effects; high assumes tech-office wages and frequent issues. The expected case is a defensible middle, not a forecast. The single most valuable improvement to this estimate is real data: headcount, crash/ANR rates and how often attendance corrections are filed (see Limitations).

Per-user formula (time-based measures)

$\text{₹/year} = \text{seconds saved per user per day} \div 3600 \times \text{loaded ₹/hour} \times \text{working days} \times N$. Example, one-tap check-in (expected): $8 \text{ s} \div 3600 \times ₹250 \times 250 \times 50 = \text{₹7k}$. Event-based measures use $\text{events per user per year} \times \text{hours each} \times \text{₹/hour} \times N$.

INPUTS

Assumptions (low / expected / high)

All inputs live in one file (`work/v030/model.py`); change a value and re-run the report to regenerate every table and chart.

INPUT	VALUE	BASIS
Employees using the app (N)	50 / 50 / 50	Placeholder. Replace with real headcount. Everything except the one-time engineering line scales linearly with N (see sensitivity chart).
Working days per year	250	Assumption (5-day weeks less holidays/leave).
Loaded employee cost / hour	₹130 / ₹250 / ₹500	Low is sourced: PLFS 2025 simple mean of male/female regular-salaried monthly earnings ₹21,285 ÷ 208 h = ₹102/h, ×1.25 overhead ^[2] . Expected/high are assumptions for office/tech staff – replace with payroll data.
Developer / IT cost / hour	₹600 / ₹1,000 / ₹1,500	Assumption (loaded cost of the person who would hot-fix and re-release).
Dev hours to fix one defect pre-release	1 / 2 / 3	Assumption, consistent with the v0.2.0–v0.3.0 fixes.
Production-vs-development fix multiplier	1.5× / 3× / 6×	Contested. The widely quoted “100× (IBM Systems Sciences Institute)” figure traces to 1981-or-older internal course notes with no published data; a 2016 study of 171 projects found resolution times usually not significantly different ^[3] . We therefore use modest multipliers.
Cost of a breached record (India)	₹6,456	IBM Cost of a Data Breach 2026, India: average ₹25.5 crore per breach and 39,500 records per breach ^[1] → ₹6,456/record. Per-record extrapolation to a small firm is an approximation.
Breachable records per employee via the app	1 / 3 / 10	Assumption (profile + contact data, attendance history, selfies).
Annual breach-probability reduction from client hardening	0.5% / 2% / 5%	Assumption. No incident data exists for this app.
Users who quit an app after crashes	5% / 15.4% / 50%	Expected = 15.4% uninstall after a single crash; high = “more than half abandon after 2–3 crashes” ^[4] (U.S. consumer survey, self-reported; employees may be more tolerant).
Exchange rate	₹96.9 = \$1	Interbank mid-market, 8 Oct 2026 ^[5] .

Benchmarks are used only where they fit: IBM's figures describe breaches at organisations in its study (average 39,500 records), so we scale by records rather than apply the ₹25.5 crore average to a 50-person firm – doing that with a 2% probability would give ₹51 lakh, which is not credible here. Survey figures on app abandonment come from U.S. consumers, not Indian employees using a mandatory tool.

What each fix avoids, and roughly what it is worth

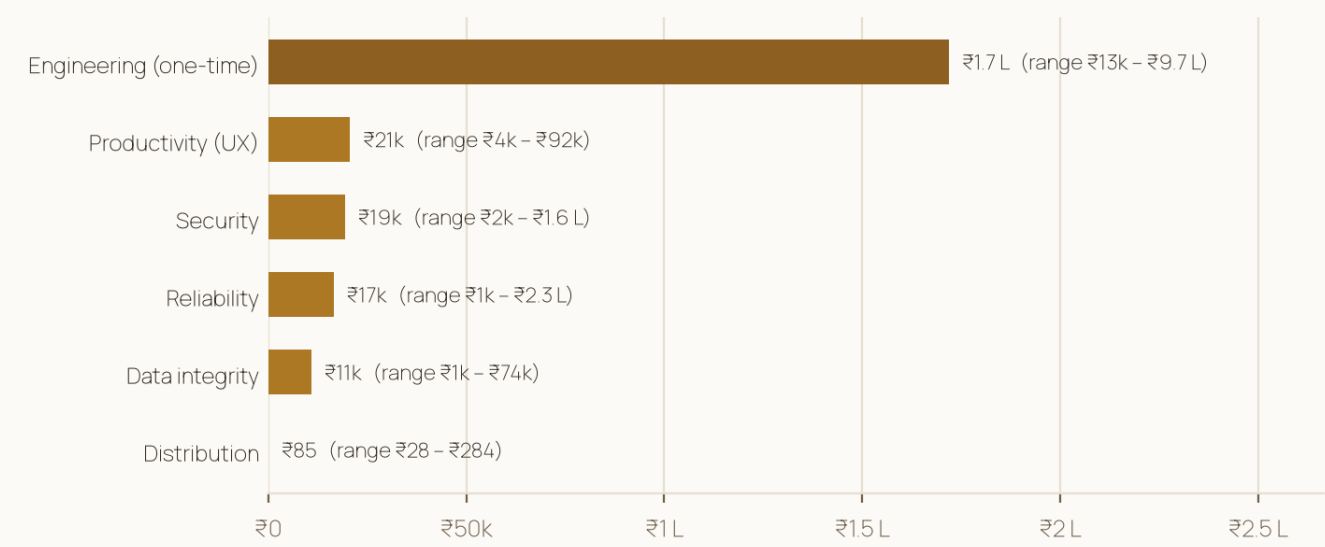
#	MEASURE	RISK / COST AVOIDED	LOW	EXPECTED	HIGH	REASONING
U1	One-tap Check in / Break / Check out on Home v0.3.0 · Productivity (UX)	Navigating to Attendance for 2–4 actions a day	₹2k	₹7k	₹26k	4/8/15 s saved per user per day
U2	Search + filters in Tasks, Directory, Leave; urgent-first task sort v0.3.0 · Productivity (UX)	Scrolling to find a task, colleague or request	₹1k	₹9k	₹43k	3/10/25 s per user per day
U3	Leave form defaults (type with balance, tomorrow) + live validation v0.3.0 · Productivity (UX)	Re-entering dates/type, rejected submissions	₹226	₹1k	₹5k	≈12 requests/user/yr × 10–60 s, spread over the year
U4	Snackbar feedback with Undo on task status; newest-first feedback v0.3.0 · Productivity (UX)	Mis-taps that need a manual re-edit; uncertainty after actions	₹451	₹3k	₹10k	≈1 mis-tap/user/month × 20–120 s
U5	Login/password: autofill, IME actions, visibility toggle, live checklist v0.3.0 · Productivity (UX)	Typing errors and failed sign-ins / password changes	₹45	₹260	₹2k	sessions persist, so sign-ins are rare (≈2/user/month)
U6	Light theme matched to web + AA contrast + 200 % font support v0.3.0 · Productivity (UX)	Mis-reading status/time; context switching web↔app	₹0	₹868	₹5k	deliberately small; mostly perceived quality, not time
R1	Crash fixes: duplicate list keys, unguarded intents, password-change crash (C-02..C-04) v0.2.0 · Reliability	App crashes in everyday flows and the time to recover	₹325	₹8k	₹1.5 L	0.25/1/3 crashes avoided per user per month × 1/3/10 min
R2	Adoption protected (users who would quit after crashes) v0.2.0 · Reliability	Falling back to the slower mobile website	₹226	₹4k	₹52k	Luciq 2026: 15.4 % uninstall after one crash, >50 % after 2–3; × 10/30/60 s/day
R3	Selfie survives process death; off-main-thread JPEG (H-06) v0.2.0 · Reliability	Lost check-ins that must be redone	₹520	₹5k	₹30k	0.2/1/3 lost selfies per user per month × 2 min
D1	Exact record ids (no silent id 0) on approvals/updates (C-01) v0.2.0 · Data integrity	Approving/updating the wrong record and the rework	₹65	₹1k	₹18k	org-wide 1/4/12 incidents per year × 0.5/1/3 h admin rework
D2	Single-flight writes, no silent POST replay (H-03, H-04) v0.2.0 · Data integrity	Duplicate check-ins, leave requests, approvals	₹433	₹4k	₹25k	0.2/1/3 duplicates per user per year × 20 min cleanup
D3	IST day bucketing for attendance/leave/calendar (M-04) v0.2.0 · Data integrity	Wrong-day records → correction requests	₹542	₹4k	₹25k	0.5/2/6 corrections avoided per user per year × 10 min (employee + manager)
D4	Confirmations for check-out / reject / sign-out (H-05) + v0.3.0 shared approve/reject guard v0.2.0 · Data integrity	Irreversible mis-taps needing admin fixes	₹325	₹2k	₹6k	0.2/0.5/1 per user per year × 15 min
S1	Client hardening: Keystore-encrypted session/cache, no backup/D2D copy, no redirects, tapjacking filter, task-affinity,	Session theft / PII exposure from a lost, shared or compromised phone	₹2k	₹19k	₹1.6 L	Δp 0.5/2/5 %/yr × users × 1/3/10 records × ₹6,456 per record (IBM 2026 India)

#	MEASURE	RISK / COST AVOIDED	LOW	EXPECTED	HIGH	REASONING
	recents privacy, response caps, log stripping v0.2.0+0.3.0 · Security					
E1	43 defects fixed before reaching users v0.2.0+0.3.0 · Engineering	Hot-fix triage, re-release and re-test cost in production	₹13k	₹1.7 L	₹9.7 L	ONE-TIME: defects × fix hours × (production multiplier – 1) × dev rate; multiplier contested
X1	APK 11.96 MB → 2.27 MB (R8 + resource shrinking) v0.2.0 · Distribution	Download data and install time per update	₹28	₹85	₹284	9.7 MB × users × updates/yr × ₹/GB — negligible in money, real for users on slow networks
Total, first year (recurring + one-time)			₹21k	₹2.4 L	₹15.3 L	≈ \$215 / \$2,472 / \$15,764

v0.2.0 IDs (C-0x, H-0x, M-0x) refer to the v0.2.0 audit. Not monetised: offline mode, Keystore session survival across restarts, design consistency with the web, and the reduced support load from clearer errors — real, but no defensible number without data.

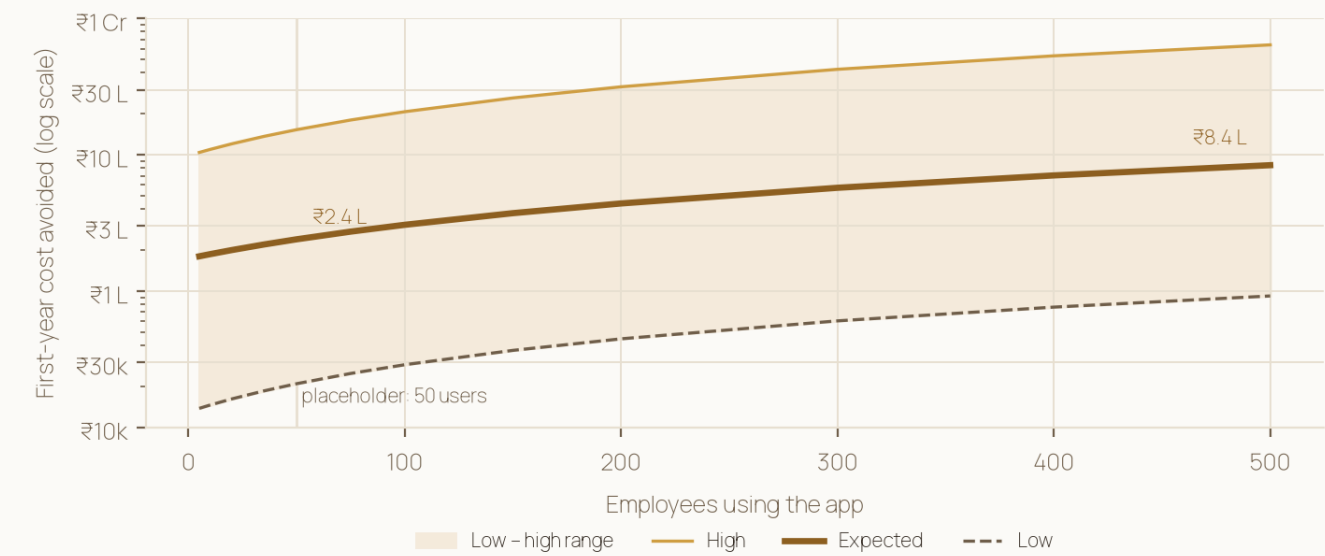
Where the value comes from

Estimated cost avoided by category — expected case, 50 users, first year



Engineering is a one-time avoidance; the other categories recur every year and scale with headcount. Distribution (APK 11.96 → 2.27 MB) is real for users on slow networks but worth only rupees in data cost.

Sensitivity: how the total moves with headcount (all other inputs fixed)



Expected total: ₹1.9 L at 10 users, ₹2.4 L at 50, ₹4.4 L at 200, ₹8.4 L at 500. The log scale keeps the wide low-high band readable.

Light theme, matched to the website

Root cause of “it shows dark”	v0.2.0 picked a dark ColorScheme from <code>isSystemInDarkTheme()</code> , and <code>enableEdgeToEdge()</code> 's default “auto” bar style turned status/nav icons white in system dark mode. Both removed.
Light only, always	Dark ColorScheme deleted; PortalTheme has no dark parameter. Window theme is <code>Theme.Material.Light.NoActionBar</code> (not <code>DayNight</code>), <code>forceDarkAllowed=false</code> , <code>windowLightStatusBar/NavigationBar=true</code> , and <code>enableEdgeToEdge(SystemBarStyle.light(...))</code> forces dark icons on parchment.
Splash & brand screens	System splash is parchment #FBFAF7 with a gold mark. Login, change-password, “can't reach Portal One” and the Home attendance card now use the web's light auth backdrop (parchment + <code>radial-gradient(circle at 100% 0, #be882e1f, transparent 42%)</code>) and a white card with the slate-200 hairline and lifted shadow — exactly the web's mobile sign-in (<code>.platform-login-card-wrap</code> / <code>.platform-login-card</code>).
Proof	<code>ThemeTest</code> launches the real activity with the device in night mode and asserts dark bar icons, a parchment window and touch filtering. Dark renders were removed from <code>UiSmokeTest</code> (one light render per screen).

Fonts — verified against the web bundle and CSS

ELEMENT	WEB CLASS	APP STYLE	TOKEN
Page titles	<code>font-display text-3xl font-semibold tracking-tight</code>	Playfair Display 600 · 30/36 · -0.025em	<code>headlineLarge</code>
Section / card h2	<code>font-display text-2xl font-semibold</code>	Playfair Display 600 · 24/32	<code>headlineMedium</code>
Card / list headings	<code>font-display text-lg font-semibold</code>	Playfair Display 600 · 18/28 (was 20 + tight tracking)	<code>headlineSmall</code>
Body	<code>body { font-family: var(--font-sans) } · slate-800</code>	Manrope 400 · 14–16	<code>body*</code>
Buttons / labels	<code>text-sm font-semibold</code>	Manrope 600 · 14	<code>labelLarge</code>
Kickers	<code>text-xs font-semibold uppercase tracking-[0.24em] text-indigo-600</code>	Manrope 600 · 12 · 0.24em (was 11 · 0.2em)	<code>Eyebrow</code>

Web loads Manrope 400–800 and Playfair Display 400–700 (Google Fonts URL in the bundle); the app ships the same variable fonts. All colours are web tokens (portal-primary #B8832B, warm slate scale, gold-remapped indigo, Tailwind status colours).

Contrast on the light palette (WCAG 2.1 AA)

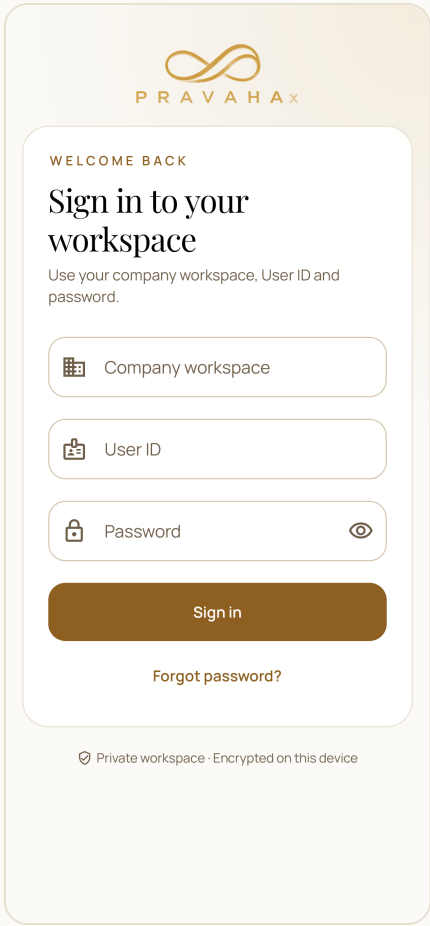
PAIR	RATIO	AA	NOTE
Filled button: white on indigo-700 #8D5F20	5.54:1	AA ✓	Web rest state is indigo-600 #AD7824 → 3.8:1 (fails AA); app uses the web's own hover token
Body text slate-800 on parchment	13.09:1	AA ✓	
Secondary text slate-600 on white	6.22:1	AA ✓	
Kicker indigo-700 on parchment	5.31:1	AA ✓	Web kicker indigo-600 → 3.7:1
Danger chip red-700 on its 10% tint	5.38:1	AA ✓	Was red-600 → 4.3:1
Success chip green-700 on its 10% tint	4.30:1	fails	
Warning chip amber-700 on its 10% tint	4.37:1	fails	
	3.83:1		Large-text threshold is 3:1

PAIR	RATIO	AA	NOTE
Check-in button label (20sp bold = large text) on #AD7824		AA ✓	

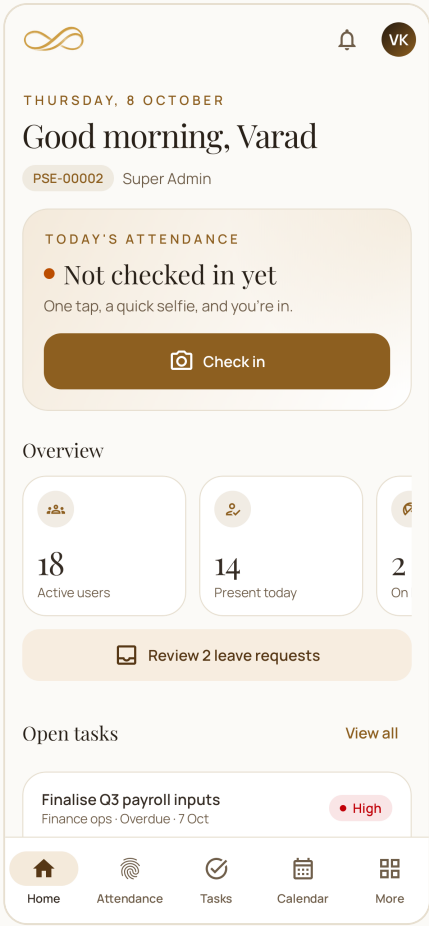
UX review, screen by screen

Kept what was already great (IST everywhere, offline banner, confirmations, skeletons, haptics). Improved the rest so each screen has one obvious primary action, feedback after every action, and fewer taps for the daily flows.

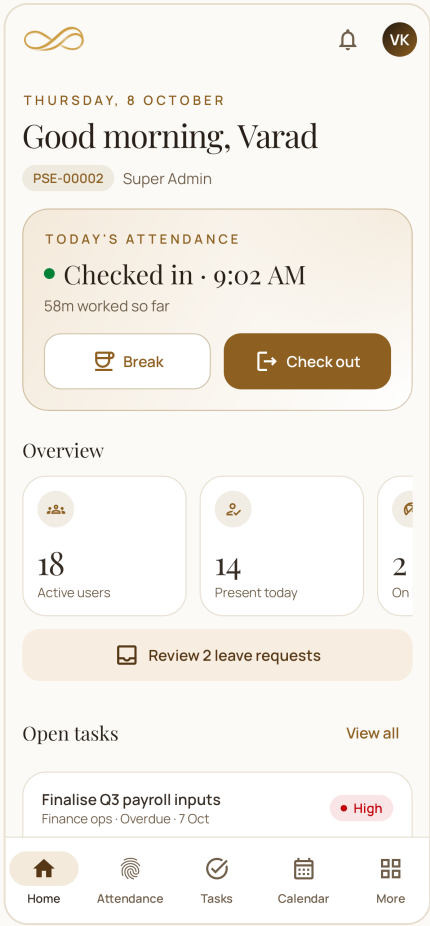
Home	Time-of-day greeting in IST; today's attendance at a glance with the one next action right there — Check in opens the selfie camera directly, then Break / Check out, or Resume. Stats show “—” instead of a fake 0 when a value is missing. “Open tasks” sorted most-urgent first with “Due today / tomorrow / Overdue”.
Attendance	Shared check-in controller (Home + tab) asks the camera for the front lens; cancelled selfie gives a clear snackbar; success messages carry the IST time; check-out confirm names the time and says a selfie follows. “This week: Xh Ym worked”. History rows say “Today”. Approve/Reject on a correction share one in-flight guard.
Tasks	Search (title, project, assignee), filters Open · Overdue · status · All with counts, urgent-first sort, swipe-to-done and status changes with Undo in the snackbar, guidance-rich empty states (Clear search / Show all).
Leave	Apply sheet pre-filled: first leave type with balance, tomorrow for one day — user only adds a reason. Status filter chips on “My requests”, keyed balance tiles, inline reason error.
Calendar, Meetings, Announcements	Day cells announce “Thursday 8 October, today, 2 events” to TalkBack; font scale capped in the fixed grid. Meetings show “Live now”, Today/Tomorrow and an accessible Join. Long announcements collapse with Read more; unread count; Mark as read gives feedback.
Directory, Profile, More	Pill search + team chips + A–Z + result count. Pull-to-refresh on More/Profile re-reads the user. New in-app Change password (Profile → Security) with live 12-character checklist. Web sections explain they open portal.pravahax.com.
Login & passwords	Light web-style card; autofill hints (username/password/new password), IME Next → Go / Send, visibility toggles, field-level error highlighting, 256-char caps, throttling after 3 failed attempts.
Everywhere	Snackbars replace stale ones; 800 ms repeat-tap guard; heading semantics; ≥48 dp targets; buttons grow with text (heightIn) for 200 % font; reduce-motion respected (static skeletons, no nav/auth transitions); predictive back enabled; stable keys on every lazy list.



Sign in – light web card

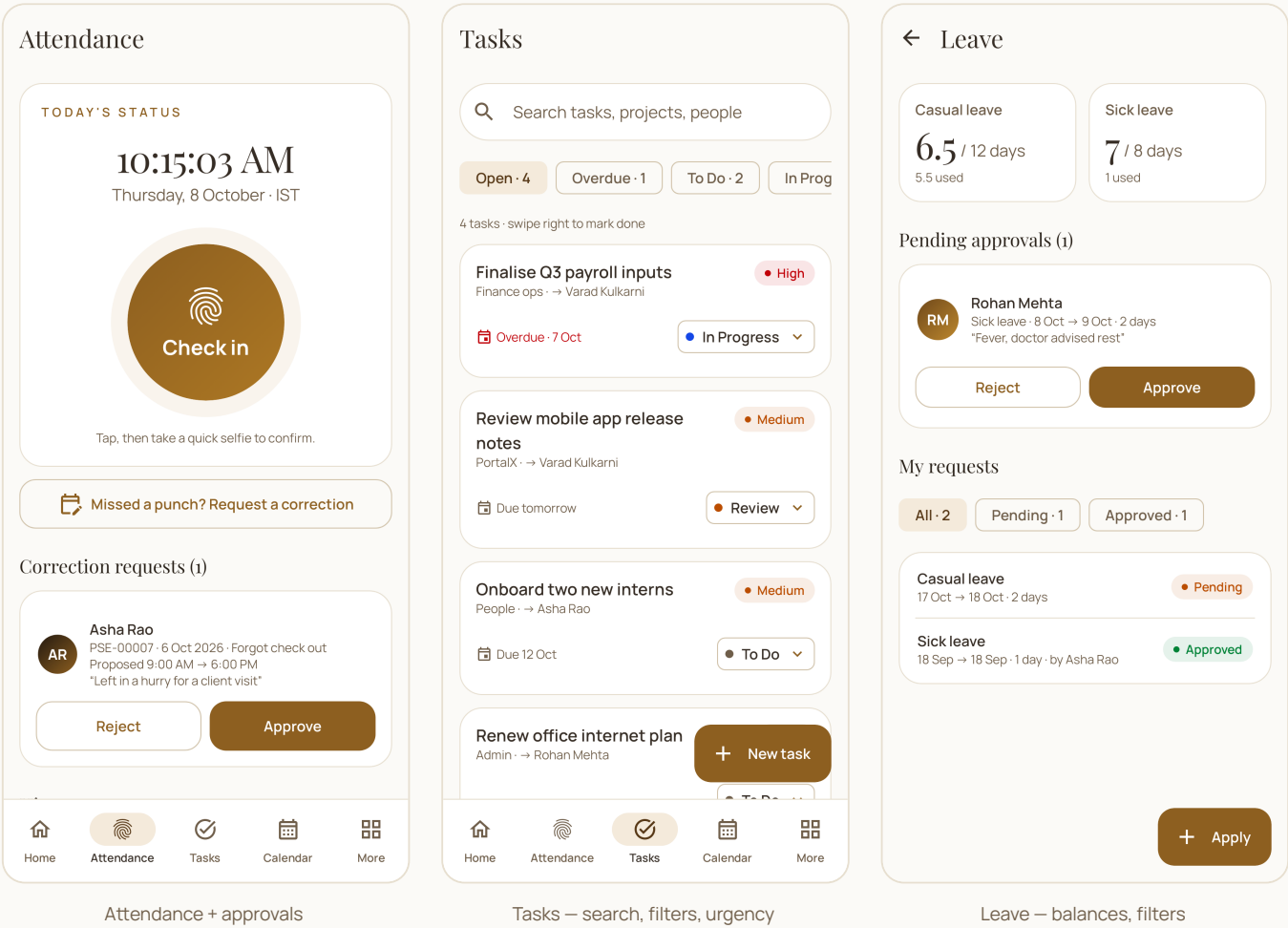


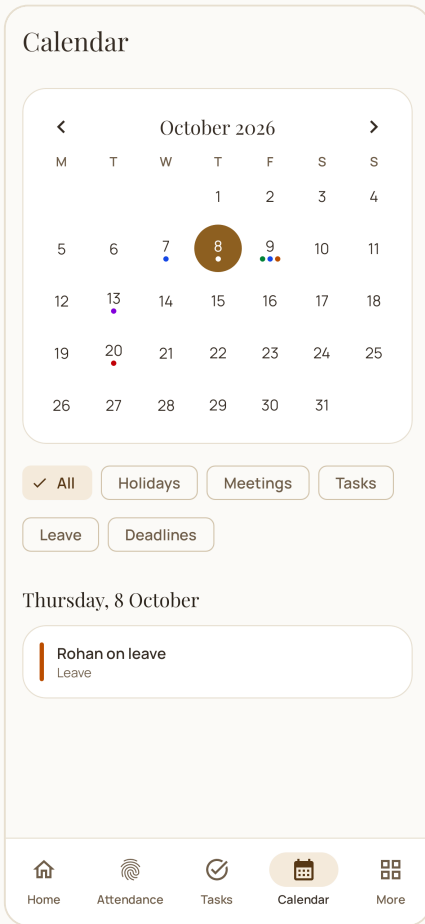
Home – not checked in: one tap



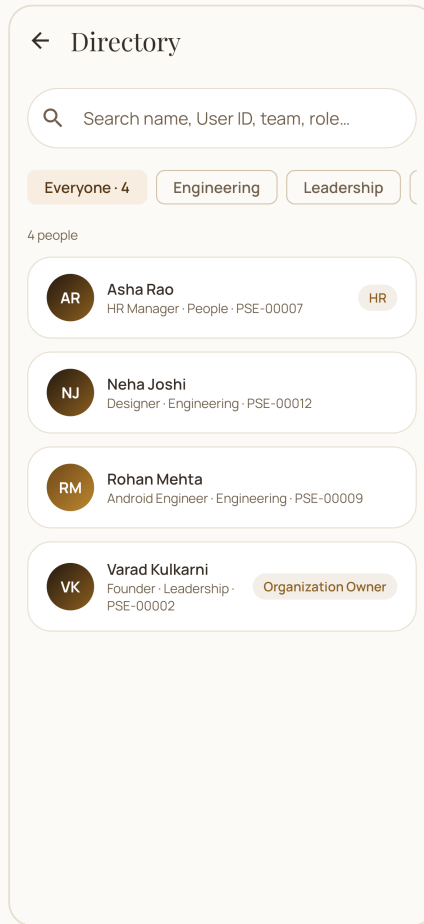
Home – working: Break / Check out

Screens (light theme, 393 × 852 dp, mock data)

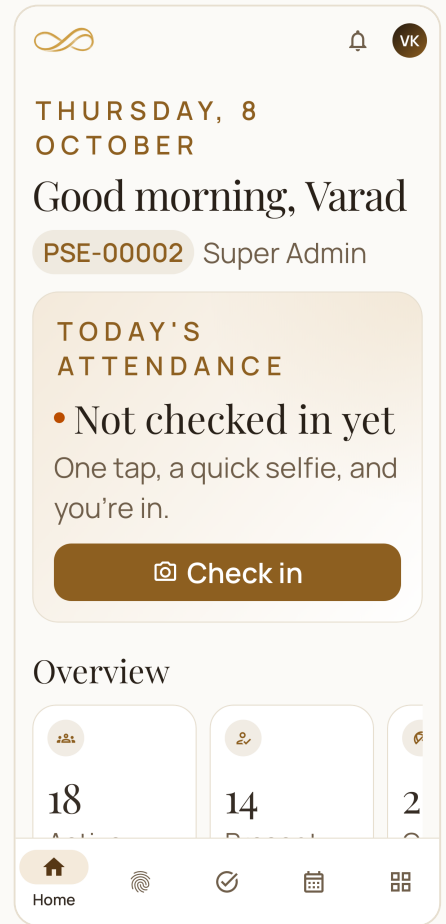




Calendar



Directory – search + teams



Home at 200 % font size

Rendered by Robolectric native graphics from the real Compose code with a mock server (names and figures are fixtures). System status/navigation bars are not part of the capture.

16 new bugs and security issues found and fixed in v0.3.0

SEV.	AREA	ISSUE	FIX
High	Security	OkHttp followed redirects, so a 3xx would replay the <code>x-portal-session-scope</code> header and the selfie POST body to the redirect target.	Redirects disabled; any 3xx is a clean error. Test: <code>doesNotFollowRedirects</code> .
High	Crash	A deeply nested JSON response overflowed the recursive JSON/seroval decoders with <code>StackOverflowError</code> (an Error, not caught) – crash.	Linear depth pre-check (≤ 256) → friendly error. Test: <code>rejectsAbsurdlyDeepJson</code> .
High	Memory	Response bodies were read without a limit (OOM on a huge/endsless body).	Bodies capped at 8 MB before decoding.
High	Theme	System dark mode produced white status/nav icons on light screens and a dark palette.	Light-only theme, forced light bar style; <code>ThemeTest</code> in night mode.
Medium	Tapjacking	Overlay apps could trick taps on Check out / Approve / Sign in.	<code>filterTouchesWhenObscured</code> on the window root (every button).
Medium	Privacy	Recents thumbnail showed attendance, directory phones/emails.	<code>setRecentsScreenshotEnabled(false)</code> (Android 13+); <code>FLAG_SECURE</code> kept on login and both password screens.
Medium	Task hijack	Default task affinity allowed StrandHogg-style activity injection.	<code>android:taskAffinity=""</code> .
Medium	Auth	Unlimited rapid sign-in attempts; unbounded password field.	Back-off after 3 failures (2 s → 30 s); 256-char cap.
Medium	Flow	Double tap could launch the camera twice (orphaned file, two camera screens).	1.5 s launch guard.
Medium	Flow	Approve and Reject on the same correction used different in-flight keys and could both be sent.	One key per correction.
Low	Flow	Repeat taps right after an action finished re-sent it.	800 ms post-completion guard per action.
Low	Data display	Home stats rendered missing values as "0".	Shows "—".
Low	A11y	Red-600 chip text on its tint was 4.3:1; gold kicker/button text under 4.5:1.	Red-700 / indigo-700 for text (see contrast table).
Low	A11y	Fixed-height buttons and the round check-in button clipped at 200 % font; tab label "Attendance" truncated.	<code>heightIn</code> everywhere, capped scale in fixed shapes, 11 sp tab labels.
Low	Logging	No guarantee against future <code>Log.d</code> of session data in release.	<code>R8 -assumenosideeffects</code> strips <code>Log.v/d/i/w/e/wtf</code> (verified: none left).
Low	UX	Snackbars queued, so feedback lagged behind actions.	Current snackbar dismissed before showing the new one.

Verification

- **Tests:** 34 tests (33 run, 1 skipped by design), 0 failures. New: `ThemeTest` (night mode), `redirect` / `deep-JSON` / `depth-scan` tests, IST relative due labels, weekly hours.
- **Release build:** R8 + resource shrinking, **2,267,706 bytes**; `versionCode` 7, `versionName` 0.3.0; `applicationId` `com.pravahax.portalx.mobile`.
- **Signature:** apksigner v2 + v3 verified, one signer, certificate SHA-256 `ad23a03f...5faf48` — same key as v0.2.0, so it installs as an update.
- **Shrunk-dex sanity check (no Robolectric):** parsed the release `classes.dex` directly — all 527 app classes present unobfuscated, manifest components present, both `META-INF/services` implementations (`coroutines Main dispatcher`)

resolve to real classes, Fn enum keeps `values()`, and no `Log.v/d/i/w/e` calls remain. The app uses no reflection-based serialization (JsonElement only), so R8 cannot break model classes.

- **Manifest:** `allowBackup=false`, data-extraction rules, cleartext off, system CAs only, FileProvider not exported, `taskAffinity=""`.

What this estimate cannot tell you — and the data that would

- **No measured usage.** Every time-saving figure is an assumption about seconds per flow. A week of anonymous screen-timing analytics would replace them.
- **No crash or ANR data.** Crash frequencies avoided are assumed. Firebase Crashlytics or Play Console Android vitals (crash-free users, ANR rate) would turn R1–R3 into measured numbers.
- **Headcount is a placeholder (50).** Multiply recurring per-user value by the real number, or read it off the sensitivity chart.
- **Wages.** Only the low case is anchored to public data (PLFS 2025); expected/high are assumptions — use actual CTC bands.
- **Security is probabilistic.** The breach-probability reduction is a judgement; IBM averages are for larger organisations and scaled here by records.
- **Contested multiplier.** The production-fix multiplier has weak empirical support; the one-time engineering line is the most uncertain.
- **Benchmarks are not Indian employees.** Abandonment figures come from a U.S. consumer survey; employees using a required tool may tolerate more.
- **Not verified on a device.** v0.3.0 was built, unit/UI-tested and dex-checked, but not installed on Varad's phone or exercised against a live account (no credentials). Server-function ids still change on every web redeploy (open item H-08 from v0.2.0).

Data that would sharpen it most

(1) real headcount and CTC bands; (2) Play Console vitals or Crashlytics for 30 days; (3) monthly count of attendance corrections and duplicate/incorrect approvals from the portal's audit log; (4) app vs. web usage share.

Sources (fetched 8 October 2026)

- [1] IBM India Newsroom — "India Records Its Highest Average Cost of a Data Breach at INR 255 Million (INR 25.5 Crore) in 2026" (3 Aug 2026)
<https://in.newsroom.ibm.com/India-Records-its-Highest-Average-Cost-of-a-Data-Breach-2026>
- [2] PIB / MoSPI — Periodic Labour Force Survey Annual Report 2025 (27 Mar 2026): regular wage/salaried avg monthly earnings ₹24,217 (male), ₹18,353 (female)
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2246009>
- [3] The Register — "Everyone cites that 'bugs are 100x more expensive to fix in production' research, but the study might not even exist" (22 Jul 2021)
<https://www.theregister.com/software/2021/07/22/everyone-cites-that-bugs-are-100x-more-expensive-to-fix-in-production-research-but-the-study-might-not-even-exist/1371906>
- [4] Luciq via Business Wire — "In 2026, Mobile Users Expect Perfection" (24 Feb 2026; survey of 1,000+ U.S. app users)
<https://www.businesswire.com/news/home/20260224897756/en/Luciq-Report-In-2026-Mobile-Users-Expect-Perfection>
- [5] BookMyForex — USD→INR interbank mid-market rate 96.9 (08-10-2026)
<https://www.bookmyforex.com/currency-converter/usd-to-inr/>